Functional / Technical

Design Document

Contoso Vendor Risk

Vendor Risk Assessment and Compliance Management Solution

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Exhibit 1
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Exhibit 2
Version history

Version	Changes	Initiator	Date
1.0	Initial Draft	Patrick Yao	31/01/2025
1.1	Updated to solution v1.0.0.2: Added Onboarding Request, Document, and Risk Scoring Config tables. Added missing fields to Vendor, Compliance Record, and Risk Assessment. Documented business rules, cloud flows, desktop flows, web resources, apps, dashboard, connection reference, relationships, option sets, and Azure infrastructure. Added missing plugin output parameters.	Patrick Yao	05/03/2026

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1 Introduction

1.1 The composition of the document

The functional design document (FDD) contains:

- Scope
- Organisation structure: The functional management of departments for creating a security model of the solution
- Business process: The schema of business process in the solution
- Architecture solution design: Object model, Data model
- Additional customisation
- Security Roles
- Integration
- Data Migration

1.2 Abbreviations

Abbreviation	Transcript
Plug-in	Plug-in is a software component that adds a specific feature to an existing software application
POC	Point of contact
FDD	Functional Design Document
TDD	Technical Design Document
API	Application Programming Interface
GUID	Globally Unique Identifier
OData	Open Data Protocol - REST-based protocol for data access
ERP	Enterprise Resource Planning
SAM	System for Award Management (SAM.gov)
OSHA	Occupational Safety and Health Administration
Virtual Table	Dataverse feature that enables external data to appear as native tables
RPA	Robotic Process Automation
PAD	Power Automate Desktop
ARM	Azure Resource Manager - infrastructure-as-code template format
IaC	Infrastructure as Code
NSG	Network Security Group
Canvas App	Lightweight Power Apps application built with a drag-and-drop designer
Model-driven App	Power Apps application driven by Dataverse data model and metadata

2 Project overview

2.1 Scope

Contoso Vendor Risk is a vendor risk management and compliance solution built on Microsoft Power Platform and Azure. It provides:

- Vendor Risk Scoring: Automated multi-factor risk assessment for vendors based on compliance status, payment behaviour, tenure, and documentation
- Bulk Compliance Checking: Ability to validate multiple vendors against compliance criteria in a single operation with fail-fast support
- ERP Integration: Real-time access to ERP vendor master data and compliance registry via OData-compliant virtual tables
- Compliance Registry Integration: SAM.gov status, debarment verification, and OSHA violation tracking
- Configurable Scoring: JSON-based configuration stored in Dataverse for weights, thresholds, and scoring rules
- Vendor Onboarding: Structured request and approval workflow for new vendor registration with eligibility checks and automatic vendor record creation
- Document Management: Upload, AI-assisted extraction, validation, and SharePoint storage of vendor compliance documents
- Copilot Integration: Three Copilot-enabled cloud flows for natural-language queries on compliance status, payment status, and onboarding requests
- RPA Processing: Power Automate Desktop flow for automated insurance certificate extraction and processing
- Model-driven App: "Vendor Intelligence Hub" providing a unified interface for vendor management, risk assessment, and compliance monitoring
- Canvas App: "VendorErpDataCard" embedded component displaying real-time ERP data on vendor forms

Technology Stack:

Layer	Technology
Data Platform	Microsoft Dataverse
Business Logic	Dataverse Plugins (C# .NET Framework 4.6.2)
Backend Services	Azure Functions (.NET 8 Isolated Worker)
External Data	Virtual Tables with OData v4 Provider
Database	Azure SQL Database
Configuration	JSON stored in contoso_apiconfig table
User Interface	Model-driven App + embedded Canvas App
Automation	Power Automate Cloud Flows + Desktop Flows (RPA)
AI	Microsoft Copilot (via cloud flow actions)
Infrastructure	Azure Resource Manager (ARM) templates
Document Storage	SharePoint Online (via connection reference)

3 Description of business processes

3.1 Common information about organisation process

This document includes the following business processes to be automated:

- Vendor Risk Assessment (Risk Score Calculation)
- Bulk Compliance Check
- ERP Data Synchronization (Virtual Table Retrieval)
- Compliance Registry Lookup
- Configuration Management
- Vendor Onboarding (Request, Eligibility Check, Approval)
- Compliance Document Processing (Upload, AI Extraction, Validation)
- Copilot-Assisted Vendor Queries (Compliance Status, Payment Status, Onboarding)
- Insurance Certificate RPA Processing
- Form Business Rules (Field Locking, Section Visibility, Data Validation)

3.2 Vendor Risk Assessment

3.2.1 Calculate Vendor Risk Score

Trigger: Custom API invocation with VendorId parameter
Purpose: Calculates a comprehensive risk score (0-100) for a vendor based on compliance, payment, tenure, and documentation factors

Process flow

Step Id	Description	Remarks
VRS-01	Validate Input	Validates VendorId is a valid GUID, AssessmentDate defaults to current UTC if not provided
VRS-02	Load Configuration	Retrieves RiskScoringConfig from contoso_apiconfig (type=100000000). Falls back to defaults if missing
VRS-03	Retrieve Vendor Data	Queries contoso_vendor for vendor details including VendorNumber, YearsInBusiness, OnboardingDate
VRS-04	Retrieve Compliance Data	Queries contoso_compliancerecord for document expiry status, counts expired/current/expiring-soon documents
VRS-05	Retrieve ERP Data	If VendorNumber exists, queries contoso_erpvendormaster (virtual table) for AvgDaysToPay, ERPStatus
VRS-06	Retrieve Registry Data	If VendorNumber exists, queries contoso_complianceregistry (virtual table) for SAMStatus, Debarred, OSHAViolations
VRS-07	Calculate Component Scores	Calculates ComplianceScore, PaymentScore, TenureScore, DocumentationScore (each 0-100)
VRS-08	Apply Weighted Calculation	Applies configurable weights (default: Compliance=0.35, Payment=0.25, Tenure=0.20, Documentation=0.20)
VRS-09	Determine Risk Category	Maps score to category: Low (>=80), Medium (>=60), High (>=40), Critical (<40 or Debarred)
VRS-10	Generate Recommendations	Creates prioritized action items based on identified risk factors
VRS-11	Return Results	Returns RiskScore, RiskCategory, RiskFactors, Recommendations, NextReviewDate, component scores

Functional design

ID	Description	Rules / Examples / Comments
VRS-FD01	Debarred vendor handling	If vendor is debarred, RiskScore forced to 0 and RiskCategory set to "Critical" regardless of other factors

VRS-FD02	Missing VendorNumber penalty	Vendors without VendorNumber receive 15-point penalty and skip ERP/Registry lookups
VRS-FD03	Compliance score deductions	Expired docs: 15 pts each (max 45). Expiring soon: 5 pts each (max 25). SAM expired: 30 pts. Debarred: 100 pts
VRS-FD04	Payment score calculation	On-time (<=30 days): 100. Slow 31-45 days: -10. Slow 46-60 days: -25. Very slow >60 days: -40
VRS-FD05	Tenure scoring	Established (>=10 yrs): 100. Mature (5-9 yrs): 85. Growing (2-4 yrs): 65. New (<2 yrs): 40
VRS-FD06	Previous assessment comparison	If previous assessment exists, calculates ScoreChange and includes in output

3.3 Bulk Compliance Check

3.3.1 Bulk Compliance Check Execution

Trigger: Custom API invocation with array of VendorIds
Purpose: Validates multiple vendors against compliance criteria in a single optimized operation

Process flow

Step Id	Description	Remarks
BCC-01	Validate Input	Parses VendorIds JSON array, validates count <= MaxVendorsPerCall (default: 100)
BCC-02	Parse Check Types	Normalizes requested check types or defaults to all: DocumentExpiry, SAMStatus, Debarment, OSHAViolations, ERPStatus
BCC-03	Load Configuration	Retrieves BulkComplianceConfig from contoso_apiconfig (type=100000001)
BCC-04	Batch Load Vendor Data	Single query retrieves all vendor records for the batch
BCC-05	Batch Load Compliance Data	Single query retrieves all compliance records with paging support (5000 per page)
BCC-06	Batch Load ERP Data	Single query to virtual table for all vendor numbers
BCC-07	Batch Load Registry Data	Single query to virtual table for all vendor numbers. If fails, sets registryFetchFailed flag
BCC-08	Process Each Vendor	Runs selected checks against cached data. If FailFast=true and vendor fails, remaining vendors are skipped
BCC-09	Determine Overall Status	Pass if all checks pass, Fail if any check fails, Warning if any warnings and no failures
BCC-10	Return Results	Returns TotalVendors, PassedCount, FailedCount, Results array, ExecutionTimeMs, FailFastTriggered

Functional design

ID	Description	Rules / Examples / Comments
BCC-FD01	DocumentExpiry check	Fail if any expired documents. Warning if expiring within 30 days or missing expiration dates
BCC-FD02	SAMStatus check	Pass if status is "Active". Fail if "Expired" or "Inactive". Warning if no data
BCC-FD03	Debarment check	Fail if debarred=true. Pass if debarred=false. Fail-closed if registry fetch failed
BCC-FD04	OSHAViolations check	Fail if count > MaxOSHAViolations (default: 3). Warning if count > OSHAWarningThreshold (default: 1)
BCC-FD05	ERPStatus check	Fail if status is "Hold" or "Inactive". Pass if "Active". Warning if no data
BCC-FD06	FailFast mode	When enabled, stops processing on first failure and reports skipped vendor IDs
BCC-FD07	Registry fail-closed	If registry batch query fails, SAMStatus, Debarment, and OSHA checks all fail (cannot bypass)

3.4 ERP Data Synchronization

3.4.1 Virtual Table Data Retrieval Flow

Trigger: Dataverse query against contoso_erpvendormaster or contoso_complianceregistry virtual tables
Purpose: Provides real-time access to external ERP and compliance registry data via OData v4 API

Process flow

Step Id	Description	Remarks
VTR-01	Query Initiation	Dataverse virtual table provider intercepts query against virtual table
VTR-02	OData Translation	Provider translates Dataverse FetchXML/QueryExpression to OData query
VTR-03	Azure Function Request	HTTP GET to Azure Functions endpoint with OData query parameters ($filter, $top, $skip)
VTR-04	SQL Query Execution	Function translates OData filter to parameterized SQL, executes against Azure SQL
VTR-05	ID Translation	Integer vendor_id/registry_id converted to deterministic GUID format for Dataverse compatibility
VTR-06	OData Response	Returns JSON with @odata.context, value array, optional @odata.count
VTR-07	Entity Mapping	Virtual table provider maps OData response to Dataverse entity format

Functional design

ID	Description	Rules / Examples / Comments
VTR-FD01	ID format conversion	Integer IDs converted to GUIDs: vendor_id=1 becomes 00000001-0000-0000-0000-000000000000
VTR-FD02	Filter support	Supports eq, ne, gt, ge, lt, le operators. String functions: contains(), startswith(), endswith()
VTR-FD03	Pagination	$top capped at 1000, default 100. $skip supported for offset pagination
VTR-FD04	SQL injection prevention	All filter values converted to parameterized queries. Column names validated against whitelist
VTR-FD05	Count support	$count=true returns total record count in @odata.count property

3.5 Vendor Onboarding

3.5.1 Onboarding Request Submission

Trigger: Manual creation of Onboarding Request record or Copilot-assisted creation via "Copilot-CreateOnboardingRequest" cloud flow
Purpose: Captures new vendor registration requests with company details, contact information, and business classification for eligibility review and approval

Process flow

Step Id	Description	Remarks
VOR-01	Create Request	User or Copilot creates contoso_OnboardingRequest with company details, contact info, business type, and primary category
VOR-02	Auto-set Submitted Date	Business rule "OnboardingRequestSetSubmittedDate" automatically stamps contoso_submitteddate on record creation
VOR-03	Eligibility Check	JS-002 (Contoso.OnboardingRequest.js) evaluates eligibility criteria and sets contoso_eligibilitypassed, populates contoso_eligibilitynotes

VOR-04	Review	Reviewer assesses request, sets contoso_reviewedby and contoso_revieweddate
VOR-05	Approve/Reject	On approval: system creates contoso_vendor record and links via contoso_convertedvendorid. On rejection: contoso_rejectionreason and contoso_rejectionnotes populated

Functional design

ID	Description	Rules / Examples / Comments
VOR-FD01	Required fields	Company Name, Contact Name, Contact Email, Business Type, and Primary Category are required for submission
VOR-FD02	Eligibility warning	JavaScript form script warns user if eligibility criteria are not met before save
VOR-FD03	Rejection visibility	Business rule shows Rejection Reason and Rejection Notes fields only when status indicates rejection
VOR-FD04	Vendor conversion	On approval, a Vendor record is created with mapped fields: CompanyName to VendorName, address fields, category, contact info

3.6 Compliance Document Processing

3.6.1 Document Upload and Validation

Trigger: Document record creation or "ProcessVendorComplianceDocument" cloud flow execution
Purpose: Manages the lifecycle of vendor compliance documents from upload through AI-assisted data extraction to validation and linking to compliance records

Process flow

Step Id	Description	Remarks
DOC-01	Document Upload	Document uploaded to SharePoint; contoso_Document record created with file metadata (filename, extension, MIME type, size) and SharePoint references
DOC-02	AI Data Extraction	"ProcessVendorComplianceDocument" cloud flow extracts structured data from document; results stored in contoso_extractionresult
DOC-03	RPA Certificate Processing	For insurance certificates, "RPAProcessInsuranceCertificate" desktop flow performs additional structured extraction
DOC-04	Link to Compliance Record	Extracted data populates or updates a contoso_ComplianceRecord via contoso_linkedcompliancerecordid lookup
DOC-05	Validation	Validator reviews extraction results, sets contoso_isvalidated, contoso_validationdate, contoso_validationnotes
DOC-06	Business Rule Check	"DocumentRequireLinkedRecord" ensures every active document has a valid parent entity link

Functional design

ID	Description	Rules / Examples / Comments
DOC-FD01	Upload source tracking	contoso_uploadsource option set records whether document came from manual upload, email, SharePoint sync, or RPA
DOC-FD02	Processing status	contoso_processedstatus tracks: Pending > Processing > Completed or Failed
DOC-FD03	Extraction errors	If AI extraction fails, contoso_extractionerrors captures error details for manual review

DOC-FD04	Coverage mapping	GL and WC coverage limits from extracted data are stored as currency fields on the linked Compliance Record

3.7 Copilot-Assisted Vendor Queries

Three Power Automate cloud flows provide natural-language Copilot interfaces:

Flow Name	Purpose	Key Data Sources
Copilot-GetVendorComplianceStatus	Returns vendor compliance status including document expiry, SAM registration, and overall risk category	contoso_vendor, contoso_compliancerecord, contoso_complianceregistry
Copilot-GetPaymentStatus	Returns vendor payment information from ERP including average days to pay, last payment, and open PO summary	contoso_vendor, contoso_erpvendormaster
Copilot-CreateOnboardingRequest	Guides user through creating a new vendor onboarding request via conversational interface	contoso_OnboardingRequest

4 Data Model

4.1 Overview

The solution uses Microsoft Dataverse as the data platform with the following core entities:

- Vendor (contoso_vendor): Master vendor records with identification, contact, and tenure data
- Compliance Record (contoso_compliancerecord): Compliance documents with expiration tracking, coverage limits, and validation status
- Risk Assessment (contoso_riskassessment): Historical risk assessment records with component scores
- API Config (contoso_apiconfig): JSON-based configuration storage
- ERP Vendor Master (contoso_erpvendormaster): Virtual table for ERP payment/financial data
- Compliance Registry (contoso_complianceregistry): Virtual table for SAM/OSHA/debarment data
- Onboarding Request (contoso_OnboardingRequest): Vendor registration requests with eligibility and approval workflow
- Document (contoso_Document): Uploaded compliance documents with AI extraction results and SharePoint references
- Risk Scoring Config (contoso_RiskScoringConfig): Named configuration records (primary config stored as JSON in API Config)

4.2 Tables

4.2.1 Vendor (contoso_vendor)

General

Option	Enable / Disable
Display Name	Vendor
Plural Name	Vendors
Schema name (Table)	contoso_vendor
Primary field (schema name)	contoso_vendorname
Auditing	Yes
Ownership	User or Team

Form (Main)

Display Name	Schema Name	Field Type	Req	Notes	Srch
Vendor ID	contoso_vendorid	GUID	+	Primary key	
Vendor Name	contoso_vendorname	String	+	Primary field	+
Vendor Number	contoso_vendornumber	String		ERP vendor identifier, used for virtual table lookups	+
Primary Category	contoso_primarycategory	OptionSet	+	Professional Services / Manufacturing / Technology / Construction / Logistics / Raw Materials / MRO Supplies / Facilities / Other	+
Years in Business	contoso_yearsinbusiness	Whole Number		Manual entry for tenure calculation	
Onboarding Date	contoso_onboardingdate	Date		Alternative tenure source if YearsInBusiness not set	
Risk Category	contoso_riskcategory	OptionSet		Low/Medium/High/Critical	+
Primary Contact	contoso_primarycontact	String		Main contact person name	
Contact Email	contoso_contactemail	String		Primary email for vendor communications	
Contact Phone	contoso_contactphone	String		Primary phone number	
Street Address	contoso_streetaddress	String		Complete street address	
City	contoso_city	String		City location	
State/Province	contoso_stateprovince	String		State or province	
Postal Code	contoso_postalcode	String		ZIP or postal code	
Country	contoso_country	String		Country of headquarters	
Website	contoso_website	String		Vendor website URL	

Tax ID/EIN	contoso_taxidein	String		Tax identification number	
Employee Count	contoso_employeecount	Whole Number		Number of employees	
Annual Revenue	contoso_annualrevenue	Currency		Reported annual revenue	
Notes	contoso_notes	Multiline Text		Additional notes about the vendor	

4.2.2 Compliance Record (contoso_compliancerecord)

General

Option	Enable / Disable
Display Name	Compliance Record
Plural Name	Compliance Records
Schema name (Table)	contoso_compliancerecord
Primary field (schema name)	contoso_name
Ownership	User or Team

Form (Main)

Display Name	Schema Name	Field Type	Req	Notes	Srch
Record ID	contoso_compliancerecordid	GUID	+	Primary key	
Name	contoso_name	String	+	Document identifier	+
Record Number	contoso_recordnumber	String		Internal reference number	
Vendor	contoso_vendor	Lookup	+	Reference to contoso_vendor	+
Compliance Doc Type	contoso_compliancedoctype	OptionSet		Insurance Certificate / SAM Registration / OSHA Record / Business License / etc.	
Certificate Number	contoso_certificatenumber	String		Certificate or document reference number	
Issue Date	contoso_issuedate	Date		Date certificate was issued	
Expiration Date	contoso_expirationdate	Date		Document expiry date	
Carrier/Issuer	contoso_carrierissuer	String		Insurance carrier or certificate issuer	
GL Coverage Limit	contoso_glcoveragelimit	Currency		General Liability coverage amount	
WC Coverage Limit	contoso_wccoveragelimit	Currency		Workers Compensation coverage amount	
Document URL	contoso_documenturl	String		URL to stored document	
Processed Status	contoso_processedstatus	OptionSet		Pending / Processing / Completed / Failed	
Processed Date	contoso_processeddate	DateTime		Date record was processed	
Extracted Data	contoso_extracteddata	Multiline Text		JSON data extracted from document	
Validation Result	contoso_validationresult	OptionSet		Pass / Fail / Warning	
Validated By	contoso_validatedby	Lookup (User)		User who performed validation	
Validated Date	contoso_validateddate	DateTime		Date validation completed	
Validation Notes	contoso_validationnotes	Multiline Text		Notes from validation review	
SharePoint Item ID	contoso_sharepointitemid	String		Document ID in SharePoint	
State	statecode	OptionSet	+	Active (0) / Inactive (1)	

4.2.3 Risk Assessment (contoso_riskassessment)

General

Option	Enable / Disable
Display Name	Risk Assessment
Plural Name	Risk Assessments
Schema name (Table)	contoso_riskassessment
Primary field (schema name)	contoso_name
Ownership	User or Team

Form (Main)

Display Name	Schema Name	Field Type	Req	Notes	Srch
Assessment ID	contoso_riskassessmentid	GUID	+	Primary key	
Name	contoso_name	String	+	Assessment reference	+
Assessment Number	contoso_assessmentnumber	String		Auto-generated reference number	
Vendor	contoso_vendorid	Lookup	+	Reference to contoso_vendor	+
Assessment Type	contoso_assessmenttype	OptionSet		Initial / Scheduled Review / Ad-hoc / etc.	+
Performed By	contoso_performedby	OptionSet		System / Manual	+
Assessment Date	contoso_assessmentdate	DateTime	+	When assessment was performed	+
Risk Score	contoso_riskscore	Decimal	+	0-100 calculated score	
Risk Category	contoso_riskcategory	OptionSet	+	Low/Medium/High/Critical	+
Compliance Score	contoso_compliancescore	Decimal		0-100 compliance component score	
Payment Score	contoso_paymentscore	Decimal		0-100 payment component score	
Tenure Score	contoso_tenurescore	Decimal		0-100 tenure component score	
Documentation Score	contoso_documentationscore	Decimal		0-100 documentation component score	
Risk Factors	contoso_riskfactors	Multiline Text		JSON array of identified risk factors	
Score Change	contoso_scorechange	Decimal		Difference from previous assessment score	
Previous Risk Score	contoso_previousriskscore	Decimal		Score from prior assessment	
Previous Risk Category	contoso_previousriskcategory	OptionSet		Risk category from prior assessment	
Recommendations	contoso_recommendations	Multiline Text		JSON array of recommended actions	
Next Review Date	contoso_nextreviewdate	DateTime		Recommended next assessment date	
Assessment Notes	contoso_assessmentnotes	Multiline Text		Detailed notes from assessment	
Triggered By Rule	contoso_triggeredbyrule	String		Which rule triggered this assessment	
State	statecode	OptionSet	+	Active (0) / Inactive (1)	

4.2.4 API Config (contoso_apiconfig)

General

Option	Enable / Disable
Display Name	API Config
Plural Name	API Configs
Schema name (Table)	contoso_apiconfig
Primary field (schema name)	contoso_name
Ownership	Organization

Form (Main)

Display Name	Schema Name	Field Type	Req	Notes	Srch
Config ID	contoso_apiconfigid	GUID	+	Primary key	
Name	contoso_name	String	+	Configuration name	+
Config Type	contoso_configtype	Whole Number	+	100000000=Risk, 100000001=Bulk	

Config JSON	contoso_configjson	Multiline Text	+	JSON configuration payload	
Is Active	contoso_isactive	Boolean	+	Only active configs are loaded	
State	statecode	OptionSet	+	Active (0) / Inactive (1)	

4.2.5 ERP Vendor Master (contoso_erpvendormaster) - Virtual Table

General

Option	Enable / Disable
Display Name	ERP Vendor Master
Plural Name	ERP Vendor Masters
Schema name (Table)	contoso_erpvendormaster
Primary field (schema name)	contoso_name
Table Type	Virtual Table (OData v4 Provider)

Form (Main)

Display Name	Schema Name	Field Type	Req	Notes	Srch
Name	contoso_name	String	+	Vendor number (maps to vendor_number)	+
Payment Terms	contoso_paymentterms	String		e.g., NET30, NET60	
Credit Limit	contoso_creditlimit	Currency		Maximum credit amount	
YTD Purchases	contoso_ytdpurchases	Currency		Year-to-date purchase value	
Open PO Count	contoso_openpocount	Whole Number		Number of open purchase orders	
Open PO Value	contoso_openpovalue	Currency		Total value of open POs	
Avg Days to Pay	contoso_avgdaystopay	Whole Number		Average payment time in days	
Last Payment Date	contoso_lastpaymentdate	DateTime		Most recent payment date	
ERP Status	contoso_erpstatus	String		Active/Hold/Inactive	

4.2.6 Compliance Registry (contoso_complianceregistry) - Virtual Table

General

Option	Enable / Disable
Display Name	Compliance Registry
Plural Name	Compliance Registries
Schema name (Table)	contoso_complianceregistry
Primary field (schema name)	contoso_name
Table Type	Virtual Table (OData v4 Provider)

Form (Main)

Display Name	Schema Name	Field Type	Req	Notes	Srch
Name	contoso_name	String	+	Vendor number	+
SAM Status	contoso_samstatus	String		Active/Expired/Inactive	
SAM Expiry	contoso_samexpiry	DateTime		SAM.gov registration expiry	
OSHA Violation Count	contoso_oshaviolationcount	Whole Number		Number of OSHA violations	
OSHA Last Inspection	contoso_oshalastinspection	DateTime		Date of last OSHA inspection	
Debarred	contoso_debarred	Boolean		True if vendor is debarred	

4.2.7 Onboarding Request (contoso_OnboardingRequest)

General

Option	Enable / Disable
Display Name	Onboarding Request

Plural Name	Onboarding Requests
Schema name (Table)	contoso_OnboardingRequest
Primary field (schema name)	contoso_requestnumber
Ownership	User or Team

Form (Main)

Display Name	Schema Name	Field Type	Req	Notes	Srch
Request ID	contoso_onboardingrequestid	GUID	+	Primary key	
Request Number	contoso_requestnumber	String		Auto-generated request identifier	+
Company Name	contoso_companyname	String	+	Legal business name	+
Contact Name	contoso_contactname	String	+	Primary contact submitting request	+
Contact Email	contoso_contactemail	String	+	Primary contact email	
Contact Phone	contoso_contactphone	String		Contact phone number	
Business Type	contoso_businesstype	OptionSet	+	Legal structure (Corporation, LLC, etc.)	
Primary Category	contoso_primarycategory	OptionSet	+	Business classification	
Street Address	contoso_streetaddress	String		Street address	
City	contoso_city	String		City	
State/Province	contoso_stateprovince	String		State or province	
Postal Code	contoso_postalcode	String		ZIP or postal code	
Country	contoso_country	String		Country	
Website	contoso_website	String		Company website URL	
Tax ID	contoso_taxid	String		Tax identification number	
Years in Business	contoso_yearsinbusiness	Whole Number		Operating years	
Employee Count Range	contoso_employeecountrange	OptionSet		Employee count bracket	
Annual Revenue Range	contoso_annualrevenuerange	OptionSet		Revenue bracket	
Source	contoso_source	OptionSet		How the request originated	
Submitted Date	contoso_submitteddate	DateTime		Auto-set on creation by business rule	
Eligibility Passed	contoso_eligibilitypassed	Boolean		Whether initial eligibility checks passed	
Eligibility Notes	contoso_eligibilitynotes	Multiline Text		Eligibility check result details	
Reviewed By	contoso_reviewedby	Lookup (User)		User who reviewed the request	
Reviewed Date	contoso_revieweddate	DateTime		Date request was reviewed	
Rejection Reason	contoso_rejectionreason	OptionSet		Reason for rejection (shown conditionally)	
Rejection Notes	contoso_rejectionnotes	Multiline Text		Additional rejection explanation	
Converted Vendor	contoso_convertedvendorid	Lookup (Vendor)		Vendor record created on approval	
State	statecode	OptionSet	+	Active (0) / Inactive (1)	

4.2.8 Document (contoso_Document)

General

Option	Enable / Disable
Display Name	Document
Plural Name	Documents
Schema name (Table)	contoso_Document

Primary field (schema name)	contoso_documentnumber
Ownership	User or Team

Form (Main)

Display Name	Schema Name	Field Type	Req	Notes	Srch
Document ID	contoso_documentid	GUID	+	Primary key	
Document Number	contoso_documentnumber	String		Reference number	+
Description	contoso_description	Multiline Text		Document description	
Document Type	contoso_documenttype	OptionSet		Insurance, Certification, License, etc.	
File Name	contoso_filename	String		Original filename	+
File Extension	contoso_fileextension	String		e.g., pdf, docx	
MIME Type	contoso_mimetype	String		File MIME type	
File Size	contoso_filesize	Whole Number		Size in bytes	
Upload Source	contoso_uploadsource	OptionSet		Manual / Email / SharePoint / RPA	
Vendor	contoso_vendorid	Lookup (Vendor)		Associated vendor	+
Onboarding Request	contoso_onboardingrequestid	Lookup (Onboarding Request)		Related onboarding request	
Linked Compliance Record	contoso_linkedcompliancerecordid	Lookup (Compliance Record)		Associated compliance record	
SharePoint URL	contoso_sharepointurl	String		Full URL in SharePoint	
SharePoint Library	contoso_sharepointlibrary	String		SharePoint library name	
SharePoint Item ID	contoso_sharepointitemid	String		SharePoint document ID	
Processed Status	contoso_processedstatus	OptionSet		Pending / Processing / Completed / Failed	
Processed Date	contoso_processeddate	DateTime		Date document was processed	
Extraction Result	contoso_extractionresult	Multiline Text		JSON from AI/ML extraction	
Extraction Errors	contoso_extractionerrors	Multiline Text		Error details from extraction	
Is Validated	contoso_isvalidated	Boolean		Whether document is validated	
Validation Date	contoso_validationdate	DateTime		Date validation was performed	
Validation Notes	contoso_validationnotes	Multiline Text		Notes from validation	
State	statecode	OptionSet	+	Active (0) / Inactive (1)	

4.2.9 Risk Scoring Config (contoso_RiskScoringConfig)

General

Option	Enable / Disable
Display Name	Risk Scoring Config
Plural Name	Risk Scoring Configs
Schema name (Table)	contoso_RiskScoringConfig
Primary field (schema name)	contoso_riskscoringconfigname
Ownership	Organization

Form (Main)

Display Name	Schema Name	Field Type	Req	Notes	Srch
Config ID	contoso_riskscoringconfigid	GUID	+	Primary key	
Name	contoso_riskscoringconfigname	String	+	Configuration name	+

Note: Primary risk scoring configuration is stored as JSON in the contoso_apiconfig table (see Section 4.2.4 and Section 5.3). This table provides named configuration records that may reference or supplement the API Config JSON.

4.3 Entity Relationships

Parent Entity	Child Entity	Relationship Type	Lookup Field
Vendor	Onboarding Request	1:N	contoso_convertedvendorid
Vendor	Risk Assessment	1:N	contoso_vendorid
Vendor	Compliance Record	1:N	contoso_vendor
Vendor	Document	1:N	contoso_vendorid
Onboarding Request	Document	1:N	contoso_onboardingrequestid
Compliance Record	Document	1:N	contoso_linkedcompliancerecordid
SystemUser	Onboarding Request (Reviewed By)	N:1	contoso_reviewedby
SystemUser	Compliance Record (Validated By)	N:1	contoso_validatedby

4.4 Global Option Sets

Schema Name	Display Name	Used By
contoso_assessmenttype	Assessment Type	Risk Assessment
contoso_businesstype	Business Type	Vendor, Onboarding Request
contoso_compliancedoctype	Compliance Doc Type	Compliance Record
contoso_employeerange	Employee Range	Onboarding Request
contoso_onboardingsource	Onboarding Source	Onboarding Request
contoso_performedby	Performed By	Risk Assessment
contoso_processedstatus	Processed Status	Compliance Record, Document
contoso_rejectionreason	Rejection Reason	Onboarding Request
contoso_revenuerange	Revenue Range	Onboarding Request
contoso_riskcategory	Risk Category	Vendor, Risk Assessment
contoso_uploadsource	Upload Source	Document
contoso_validationresult	Validation Result	Compliance Record
contoso_vendorcategory	Vendor Category	Vendor (Primary Category)

5 Additional functionality

5.1 Plugin - CalculateVendorRiskScorePlugin

A custom Dataverse plugin registered as a Custom API that calculates comprehensive risk scores for vendors.

Plugin Registration:

Property	Value
Assembly	Contoso.VendorRisk.Plugins.dll
Class	Contoso.VendorRisk.Plugins.Plugins.CalculateVendorRiskScorePlugin
Message	contoso_CalculateVendorRiskScore (Custom API)
Stage	Main Operation

Input Parameters:

Parameter	Type	Required	Description
VendorId	GUID	Yes	ID of the vendor to assess
AssessmentDate	DateTime	No	Date for assessment (defaults to UTC now)
IncludeHistoricalTrend	Boolean	No	Whether to include previous assessment comparison

Output Parameters:

Parameter	Type	Description
RiskScore	Decimal	Overall risk score (0-100)
RiskCategory	String	Low/Medium/High/Critical
RiskFactors	String	JSON array of risk factors
ComplianceScore	Decimal	Component score for compliance
PaymentScore	Decimal	Component score for payment behaviour
TenureScore	Decimal	Component score for business tenure
DocumentationScore	Decimal	Component score for documentation completeness
Recommendations	String	JSON array of recommended actions
NextReviewDate	DateTime	Recommended next review date
PreviousRiskScore	Decimal	Risk score from the most recent prior assessment (0 if none exists)
PreviousRiskCategory	String	Risk category from prior assessment (empty string if none)
ScoreChange	Decimal	Difference between current and previous risk score (0 if no prior)
ConfigSource	String	Where configuration was loaded from: "Database" or "Defaults"
ConfigWarning	String	Warning message if configuration issue was detected (empty if OK)

5.2 Plugin - BulkComplianceCheckPlugin

A custom Dataverse plugin for batch compliance validation of multiple vendors.

Plugin Registration:

Property	Value
Assembly	Contoso.VendorRisk.Plugins.dll
Class	Contoso.VendorRisk.Plugins.Plugins.BulkComplianceCheckPlugin
Message	contoso_BulkComplianceCheck (Custom API)
Stage	Main Operation

Input Parameters:

Parameter	Type	Required	Description
VendorIds	String	Yes	JSON array of vendor GUIDs

CheckTypes	String	No	JSON array of check types to run (defaults to all)
FailFast	Boolean	No	Stop processing on first failure

Output Parameters:

Parameter	Type	Description
TotalVendors	Integer	Number of vendors requested
ProcessedCount	Integer	Number of vendors actually processed
PassedCount	Integer	Number of vendors that passed all checks
FailedCount	Integer	Number of vendors that failed at least one check
Results	String	JSON array of VendorComplianceResult objects
ExecutionTimeMs	Integer	Total execution time in milliseconds
FailFastTriggered	Boolean	Whether FailFast mode stopped processing
RequestedCount	Integer	Number of vendor IDs in the original request
SkippedVendorIds	String	JSON array of vendor GUIDs skipped due to FailFast
ConfigSource	String	Where configuration was loaded from: "Database" or "Defaults"
ConfigWarning	String	Warning message if configuration issue was detected (empty if OK)

5.3 Risk Scoring Configuration

JSON configuration stored in contoso_apiconfig allows customization of risk scoring behaviour without code changes.

Configuration Structure (RiskScoringConfig):

Section	Description
Weights	Compliance: 0.35, Payment: 0.25, Tenure: 0.20, Documentation: 0.20
Thresholds	Low: 80, Medium: 60, High: 40
Deductions	ExpiredDocumentPer: 15, Debarred: 100, SamExpired: 30, OshaViolationPer: 5, etc.
ReviewDays	Critical: 7, High: 30, Medium: 90, Low: 180
TenureScores	Established (>=10 yrs): 100, Mature (5-9): 85, Growing (2-4): 65, New (<2): 40
DocumentExpiryWarningDays	30 days warning threshold

5.4 Business Rules

Classic workflows (real-time) that enforce form behaviour and data integrity:

#	Rule Name	Entity	Trigger	Behaviour
1	VendorLockFieldsWhenSuspended	Vendor	On status change	When vendor status is set to Suspended, locks editable fields to prevent modification
2	ComplianceShowCoverageSection	Compliance Record	On form load / field change	Dynamically shows or hides the coverage information section (GL/WC limits) based on Compliance Doc Type
3	OnboardingShowRejectionSection	Onboarding Request	On status change	Displays Rejection Reason and Rejection Notes fields only when the request is rejected
4	RiskLockSystemGeneratedFields	Risk Assessment	On form load	When Performed By = "System", locks 10 calculated fields: Risk Score, Risk Category, Compliance Score, Payment Score, Tenure Score, Documentation

				Score, Risk Factors, Score Change, Previous Risk Score, Previous Risk Category
5	DocumentRequireLinkedRecord	Document	On save	Validates that every Document record has at least one parent entity link (Vendor or Compliance Record)
6	OnboardingRequestSetSubmittedDate	Onboarding Request	On create	Automatically sets contoso_submitteddate to the current date/time when a new request is created

5.5 Web Resources

JavaScript libraries registered on entity forms:

Web Resource	Entity	Purpose
contoso_/scripts/Contoso.Common.js	(Shared)	Reusable utility functions for form scripts across all entities
contoso_/scripts/Contoso.Vendor.js	Vendor	JS-001: High Risk Notification — displays a notification banner when the vendor's risk category is High or Critical
contoso_/scripts/Contoso.OnboardingRequest.js	Onboarding Request	JS-002: Eligibility Warning — evaluates eligibility criteria on form load/change and warns the user if criteria are not met

5.6 Power Automate Cloud Flows

Flow Name	Type	Trigger	Purpose
ProcessVendorComplianceDocument	Automated	Document record creation	Orchestrates AI-assisted data extraction from uploaded compliance documents, populates extraction results, and updates processing status
Copilot-GetVendorComplianceStatus	Copilot-enabled	Natural language query	Returns vendor compliance status including document counts, expiry status, SAM registration, and risk category
Copilot-CreateOnboardingRequest	Copilot-enabled	Natural language request	Guides conversational creation of a new Onboarding Request record with required fields
Copilot-GetPaymentStatus	Copilot-enabled	Natural language query	Returns vendor payment data from ERP virtual table including average days to pay, last payment, and open PO summary

Note: APITESTFLOW and APITESTCOMPLIANCE are development/test flows for API validation and are not intended for production use.

5.7 Desktop Flows (RPA)

Flow Name	Purpose	Infrastructure
RPAProcessInsuranceCertificate	Automates extraction of structured data from insurance certificate documents (PDF/image). Extracts carrier name, certificate number, issue/expiration dates, and coverage limits. Results are written back to the Compliance Record entity.	Runs on PAD_Demo_VM (Azure VM with Power Automate Desktop agent)

5.8 Applications

5.8.1 Model-Driven App — Vendor Intelligence Hub

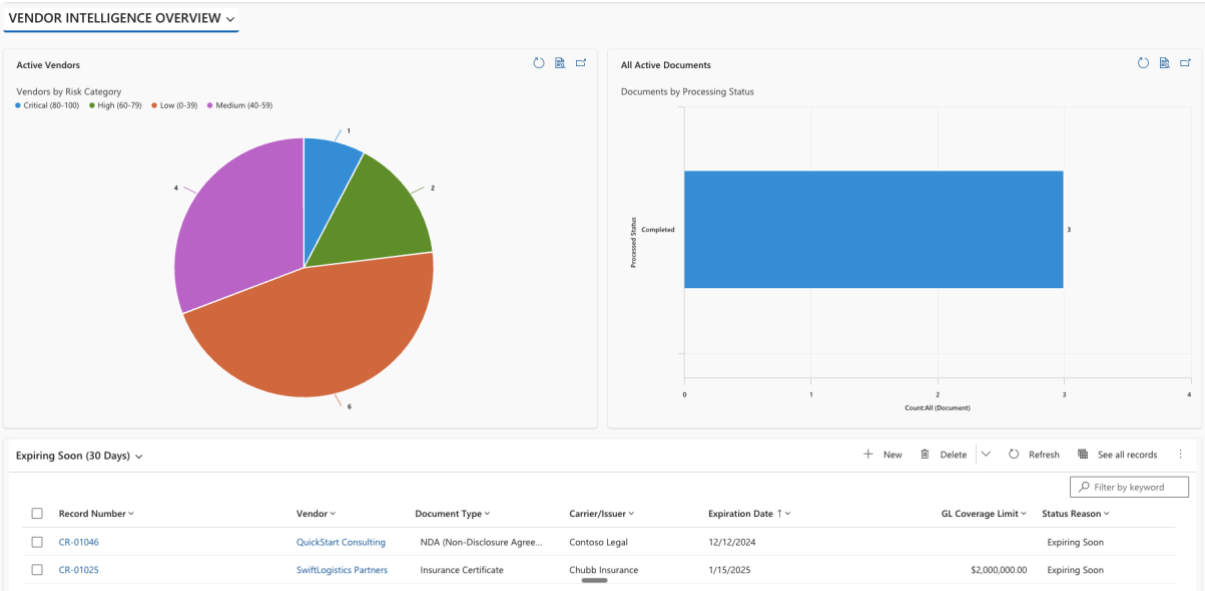
The primary user interface for the solution. Provides navigation to all entities via a unified sitemap including:
Vendor management (list and detail views)
Compliance Record tracking
Risk Assessment history
Onboarding Request workflow
Document management
API configuration and risk scoring settings

5.8.2 Canvas App — VendorErpDataCard

An embedded canvas app component (contoso_VendorErpDataCard) displayed on the Vendor main form. Connects to the ERP Vendor Master virtual table and renders real-time financial data including payment terms, credit limit, YTD purchases, open PO count/value, and average days to pay.

5.9 Dashboard — Compliance Status Distribution

A system dashboard that visualises compliance status distribution across the vendor base. Aggregates compliance records by status code and displays record counts to provide a quick overview of overall compliance health.



6 Integration

6.1 Logging mechanism

Plugin execution traces are logged via ITracingService for debugging in Dataverse plugin trace logs.
Azure Function errors and informational logs are written via ILogger to Application Insights when configured.

6.2 Integration 1 - ERP OData API (VendorMaster)

Provides OData v4-compliant access to ERP vendor master data for Dataverse virtual tables.

6.2.1 Scope

Called by Dataverse virtual table provider when queries are made against contoso_erpvendormaster.

6.2.2 Interface design

Overview

Property	Value
Endpoint	GET /api/odata/VendorMaster
Single Entity	GET /api/odata/VendorMaster({guid})
Metadata	GET /api/odata/$metadata
Runtime	Azure Functions .NET 8 Isolated
Hosting	Consumption Plan
Database	Azure SQL Database

Request Format
OData query parameters: $filter, $top (max 1000, default 100), $skip, $count
Response Format
OData v4 JSON with @odata.context, @odata.count (if requested), value array containing VendorMaster entities.

6.3 Integration 2 - ERP OData API (ComplianceRegistry)

Provides OData v4-compliant access to compliance registry data (SAM.gov, OSHA, debarment).

6.3.1 Scope

Called by Dataverse virtual table provider when queries are made against contoso_complianceregistry.

6.3.2 Interface design

Overview

Property	Value
Endpoint	GET /api/odata/ComplianceRegistry
Single Entity	GET /api/odata/ComplianceRegistry({guid})
Runtime	Azure Functions .NET 8 Isolated

Request Format
OData query parameters: $filter (supports vendor_number, sam_status, debarred, osha_violation_count), $top, $skip, $count
Response Format
OData v4 JSON with ComplianceRegistry entities including sam_status, sam_expiry, osha_violation_count, osha_last_inspection, debarred.

6.4 Environment Variables

Schema Name	Description
SqlConnectionString	Azure SQL Database connection string for ERP data
AzureWebJobsStorage	Azure Storage connection for Functions runtime
FUNCTIONS_WORKER_RUNTIME	Set to "dotnet-isolated" for .NET 8 isolated worker
APPLICATIONINSIGHTS_CONNECTION_STRING	Application Insights connection for Function App telemetry

6.5 Connection Reference

Property	Value
Name	Contoso ERP OData
Connector Type	OData v4 Data Provider
Purpose	Provides the connection for virtual table integration between Dataverse and the Azure Functions OData API
Used By	contoso_erpvendormaster, contoso_complianceregistry
Dependency	OData v4 Data Provider package (v9.1.0.445)

6.6 Azure Infrastructure

Deployed via ARM template (Infrastructure/rg-contoso-vendor-demo-template.json):

Resource Type	Name	Purpose
Function App	func_contoso_erp_api	Hosts OData API endpoints (Consumption tier)
App Service Plan	AustraliaEastPlan	Consumption (Y1 Dynamic) plan for Function App
Application Insights	func_contoso_erp_api	Telemetry and monitoring for Function App
SQL Server	sql_contoso_demo_*	Hosts ERP data (TLS 1.2 minimum, Azure AD auth)
Storage Account	stcontosoerpfunc	Function App runtime storage (Standard_LRS, HTTPS only)
Virtual Machine	PAD_Demo_VM	Hosts Power Automate Desktop agent for RPA flows
Virtual Network	PAD_Demo_VMVNET	Network isolation for VM (10.0.0.0/16)
Network Security Group	PAD_Demo_VMNSG	Firewall rules for VM (RDP 3389 inbound)
Public IP Address	PAD_Demo_VMPublicIP	Static public IP for VM access